


nvidia/gpt-oss-120b-Eagle3-throughput

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 Eagle3

 License: nvidia-open-model-license

⋮

 **Model card**
 Files

 xet

 Community 3

Model Overview

Description:

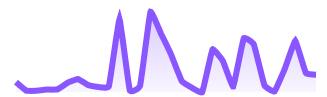
The NVIDIA gpt-oss-120b Eagle model is the Eagle head of the OpenAI's gpt-oss-120b model, which is an auto-regressive language model that uses a mixture-of-experts (MoE) architecture with 5 billion activated parameters and 120 billion total parameters. For more information, please check [here](#). The NVIDIA gpt-oss-120b Eagle3 model incorporates Eagle speculative decoding with [Model Optimizer](#).

This model is ready for commercial/non-commercial use.

Note

This EAGLE3 Module is only usable for drafting a single predicted token. It has high acceptance rate and is useful for high-concurrency inference where a single speculated token is the optimal configuration. It also retains strong acceptance rate for long-context usage.

Downloads last month
1,034



⚡ Inference Providers NEW

 Text Generation

This model isn't deployed by any Inference Provider.


[Ask for provider support](#)

🌳 Model tree for nvidia/gpt-oss-12...

Base model [openai/gpt-oss-120b](#)

🧱 **Finetuned** (105) [this model](#)

📁 Collection including nvidia/gpt-o...

Speculative Decoding ...  Collection

A collection ... • 16 items • U... • △ 12

License/Terms of Use:

nvidia-open-model-license Apache License 2.0

Deployment Geography:

Global

Use Case:

Developers designing AI Agent systems, chatbots, RAG systems, and other AI-powered applications. Also suitable for typical instruction-following tasks.

Release Date:

Huggingface: Aug 20th, 2025 via
[<https://huggingface.co/nvidia/gpt-oss-120b-Eagle3>].

Model Architecture:

Architecture Type: Transformers

Network Architecture: gpt-oss-120b

##Computational Load **Cumulative Compute:
4.8x10²⁰ **Estimated Energy and Emissions for
Model Training: *Total kWh = 2500 *Total Emissions
(tCO₂e) = 0.8075

Input:

Input Type(s): Text

Input Format(s): String

Input Parameters: One Dimensional (1D):
Sequences

Output:

Output Type(s): Text

Output Format: String

Output Parameters: One-Dimensional (1D):

Sequences

Our AI models are designed and/or optimized to run on NVIDIA GPU-accelerated systems. By leveraging NVIDIA's hardware (e.g. GPU cores) and software frameworks (e.g., CUDA libraries), the model achieves faster training and inference times compared to CPU-only solutions.

Software Integration:

Supported Runtime Engine(s):

- TensorRT-LLM

Supported Hardware Microarchitecture

Compatibility:

- NVIDIA Blackwell

Preferred Operating System(s):

- Linux

The integration of foundation and fine-tuned models into AI systems requires additional testing using use-case-specific data to ensure safe and effective deployment. Following the V-model methodology, iterative testing and validation at both unit and system levels are essential to mitigate risks, meet technical and functional requirements, and ensure compliance with safety and ethical standards before deployment.

Model Version(s):

** The model is quantized with nvidia-modelopt

v0.35.0

Training and Evaluation Datasets:

** The total size (in number of data points) 503.3K

** Total number of datasets 2

** Dataset partition: Training 100%

Training Dataset:

Link: [ultrachat_200k](#) and [nemotron post-training v2 dataset](#), only prompts from the datasets were used for data synthesis (the original responses from GPT were not used for data synthesis) which is then used to train the Eagle modules. Click the links above for more information regarding the dataset.

** Data Modality [Text]

** Data Collection Method by dataset

- Hybrid: Synthetic, Human, Automated

** Labeling Method by dataset

- Hybrid: Synthetic, Human, Automated

Properties: 1.6M samples, majority synthetic, others sourced from commercially-friendly datasets.

Evaluation Dataset:

Link: MTBench, for more details, see [here](#)

**** Data Collection Method by dataset**

- Hybrid: Human, Synthetic

**** Labeling Method by dataset**

- Hybrid: Human, Synthetic

Properties: 3,300 multi-turn dialogue sequences, each annotated with expert preference votes.

Inference:

Engine: TensorRT-LLM 1.1.0rc1

Test Hardware: B200

Eagle Speculative Decoding

Synthesized data was obtained from OpenAI's gpt-oss-120b model, which is then used to finetune the Eagle modules. This model is ready for inference with TensorRT-LLM in Eagle speculative decoding mode. Eagle modules are used to predict candidate tokens beyond the next token. In the generation step, each forward Eagle module generates a distribution of tokens beyond the previous. Then, a tree-based attention mechanism samples some candidate sequences for the original model to validate. The longest accepted candidate sequence is selected so that more than 1 token is returned in the generation step. The number of tokens generated in each step is called acceptance rate.

Usage

To serve the checkpoint with [TensorRT-LLM](#), follow the sample commands below with the TensorRT-

LLM GitHub repo:

```
trtllm-serve <gpt-oss-120b checkpoint> --hos
```

extra-llm-api-config.yml is like this

```
enable_attention_dp: false
disable_overlap_scheduler: true
enable_autotuner: false

cuda_graph_config:
  max_batch_size: 1

speculative_config:
  decoding_type: Eagle
  max_draft_len: 3
  speculative_model_dir: <eagle3 checkpoint>

kv_cache_config:
  enable_block_reuse: false
```

Evaluation

The Eagle acceptance rate benchmark results (MT-Bench) with draft length 1 are presented in the table below for medium reasoning

It has higher acceptance rate at single predicted token.

Category	MT Bench Acceptance Rate
writing	1.72
roleplay	1.66
reasoning	1.72

Category	MT Bench Acceptance Rate
math	1.84
coding	1.79
extraction	1.80
stem	1.70
humanities	1.64

Ethical Considerations

NVIDIA believes Trustworthy AI is a shared responsibility and we have established policies and practices to enable development for a wide array of AI applications. When downloaded or used in accordance with our terms of service, developers should work with their internal model team to ensure this model meets requirements for the relevant industry and use case and addresses unforeseen product misuse.

For more detailed information on ethical considerations for this model, please see the Model Card++ Explainability, Bias, Safety & Security, and Privacy Subcards.

Please report security vulnerabilities or NVIDIA AI Concerns [here](#). SUBCARDS:

Explainability

Field:	Response:
Intended Application(s) &	Text generation, reasoning, summarization, and question

Field:	Response:
Domain(s):	answering.
Model Type:	Text and Image-to-text transformer
Intended Users:	This model is intended for developers, researchers, and customers building/utilizing LLMs, while balancing accuracy and efficiency.
Output:	Text String(s)
Describe how the model works:	Generates text by predicting the next word or token based on the context provided in the input sequence using multiple self-attention layers
Technical Limitations:	The model was trained on data that contains toxic language and societal biases originally crawled from the internet. Therefore, the model may amplify those biases and return toxic responses especially when prompted with toxic prompts. Therefore, before deploying any applications of this model, developers should perform safety testing and tuning tailored to their specific applications of the model.
Verified to have met prescribed quality standards?	Yes
Performance Metrics:	Accuracy, Throughput, and user-side throughput
Potential Known Risk	The model may generate answers that may be inaccurate, omit key information, or include irrelevant or redundant text producing socially unacceptable or undesirable text, even

Field:	Response:
	if the prompt itself does not include anything explicitly offensive.
Licensing:	Your usage is governed by the following license

Bias

Field:	Response:
Participation considerations from adversely impacted groups (protected classes) in model design and testing:	None
Measures taken to mitigate against unwanted bias:	None

Safety & Security

Field:	Response:
Model Application(s):	Chat, Instruction Following, Chatbot Development, Code Generation, Reasoning
Describe life critical application (if present):	None Known
Use Case Restrictions:	Abide by the license
Model and Dataset Restrictions:	The Principle of least privilege (PoLP) is applied limiting access for dataset generation. Restrictions enforce dataset access during training, and dataset license constraints adhered

Field:	Response:
	to. Model checkpoints are made available on Hugging Face, and may become available on cloud providers' model catalog.

Privacy

Field:	Response:
Generatable or Reverse engineerable personal data?	None
Was consent obtained for any personal data used?	None Known
Personal data used to create this model?	None Known
How often is dataset reviewed?	Before Release
Is there provenance for all datasets used in training?	Yes
Does data labeling (annotation, metadata) comply with privacy laws?	Yes